

GADING ADITYA PERDANA

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Central Jakarta, Indonesia

SUMMARY

First-generation undergraduate researcher and engineer. Author of five peer-reviewed publications (four first-author) with a focus on computer vision and deep learning. Currently developing on-device AI systems for medical imaging and conducting research on compact vision transformers. Experienced in PyTorch, full-stack development, and deploying machine learning models to production. Self-funding education through freelance work while maintaining a full-time academic load.

EDUCATION

Binus University | GPA: 3.63/4.00

Jakarta, Indonesia

Bachelor of Science in Computer Science, Specialization in Intelligence Systems

Aug. 2023 – Feb 2027

- Accelerated 3.5-year program; **Apple Developer Academy Scholar (Cohort 2026)**
- Research focus: Medical Imaging, Vision Transformers, Efficient AI, and Machine Learning Operations.

PUBLICATIONS

- **Perdana, G. A.**, Nasari, M., Saputra, M.A., Halim, R., Minor, K.A. "Enhancing Bone Fracture Detection in X-ray Images Using Faster R-CNN with EfficientNetV2." *IEEE ISRITI* (2025). **First author.**
- **Perdana, G. A.**, Ghazali, M.A., Iswanto, I.A., Joddy, S. "CALM: Calibrated Adaptive Learning via Mutual-Ensemble Fusion." *Procedia Computer Science* (ICCCSCI 2025). **First author.**
- **Perdana, G. A.**, Kisdi, M.I.A., Wairooy, I.K., Makalew, B.A. "Analytical Analysis of Cryptocurrency Regulation and Adoption: A Machine Learning-Driven Ablation Study of the United States, Russia, and Indonesia." *Procedia Computer Science* (ICCCSCI 2025). **First author.**
- Nathen, A., **Perdana, G. A.**, Jefferson, G., Hasani, M.F., Maulina, A., Tjahyadi, B.G. "Tiny vs. Tinier: Baseline ViT-Tiny vs. Ensemble-Distilled Student on Imbalanced Fracture Detection." *Procedia Computer Science* (ICCCSCI 2025).
- **Perdana, G. A.**, Wijaya, I.I., Fahreza, K.A., Tarigan, G.A. "Ablation Study: Calibrated Adaptive Learning Ensemble Methodology." *Procedia Computer Science* (ICCCSCI 2025). **First author.**

AWARDS & HONORS

Best Presenter Award

Semarang, Indonesia

The 10th International Conference on Computer Science (ICCCSCI)

August 2025

- Awarded for technical communication and research defense among international doctoral and academic presenters.

EXPERIENCE

Apple Developer Academy Scholar

Feb. 2026 – Present

Apple & Binus University

BSD City, Indonesia

- Selected for the 2026 cohort to engineer native iOS/macOS applications using Swift, SwiftUI, and CoreML.
- Collaborating within cross-functional teams employing Challenge-Based Learning to design, build, and ship production-ready applications to the App Store.

Undergraduate Research Assistant

Jan 2025 – Dec 2025

Binus University (Bina Nusantara University)

Anggrek Campus, Jakarta

- **Lead Author & Researcher:** Led experimental design for bone fracture detection, progressing from research assistant to first authorship on an IEEE publication.
- **Data Engineering:** Designed a preprocessing pipeline to mitigate severe class imbalance in X-ray data.
- **Model Architecture:** Benchmarked FPN vs. PAFPN architectures; engineered a Faster R-CNN model with an EfficientNetV2-S backbone, implementing cost-sensitive loss functions.
- **Training Pipeline:** Optimized training using AdamW, mixed precision (AMP), and Exponential Moving Average (EMA); achieved a 21.6% mAP improvement over baseline models.

Freelance Software Developer & Independent Tutor

Aug. 2021 – Jan. 2026

Self-Employed

Remote & Jakarta, Indonesia

- Various contract work including full-stack development, research consulting, and courier/logistics services, demonstrating strong self-management and client communication skills.
- Delivered technical projects ranging from web applications to research (Computer Vision) methodology consultation for diverse clients.

SELECTED PROJECTS

OccamViT: A Subtractive Audit of Compact Vision Transformers	Aug. 2025 – Present
- Architected and trained from scratch a minimalist isotropic ViT family reaching 80.2% top-1 at 6.8M params (Tiny) and 82.1% at 11.9M (Small) on ImageNet-1K with no distillation or extra data, exceeding DeiT-S accuracy at 3.3× fewer parameters. Paper in preparation for CVPR 2027 (First Author). - Conducted a noise-calibrated subtractive audit ($\sigma \approx 0.31$) systematically dismantling a 16.6M 4-attention prototype; proved complex attention modules (XCA, token banks, landmark pooling) are non-essential, isolating the inductive gains to a 3-stage convolutional trunk (−4.3% when removed) and depthwise FFN mixing. - Discovered emergent zero-shot robustness with OccamViT-S achieving 31.7% on ImageNet-A (+6.0% over matched peers) without specialized objectives; audited published compact ViT baselines to uncover that mixed evaluation lanes flip 17 of 78 leaderboard rankings. Built in pure PyTorch (no custom CUDA kernels) on a single RTX 5090.	
DALX: On-Device AI Chest Radiograph Measurement Platform	July. 2026 – August 2026
- Engineered an on-device biomedical imaging system and native macOS SwiftUI app executing 5 local deep learning models (HybridGNet, CXAS, RAD-DINO, MedGemma) to extract deterministic anatomical measurements - Selected to exhibit at Garuda Spark Expo, demonstrating real-time, zero-cloud DICOM processing, automated contouring, and structured DICOM SR/PDF export to industry leaders and healthcare practitioners. - Secured clinical validation interest from Universitas Indonesia Faculty of Medicine (FK UI – Occupational Medicine Division) to explore automated occupational lung disease screening workflows, including potential development for ILO pneumoconiosis and lung cancer detection.	
CALM: Calibrated Adaptive Learning via Mutual-Ensemble Fusion	Feb. 2025 – May 2025
- Designed a novel ensemble framework integrating knowledge distillation, mutual learning, and calibration to enhance model confidence and reliability. - Introduced adaptive curriculum protocol to dynamically schedule training objectives, optimizing multi-teacher fusion. - Achieved 97.16% accuracy on CIFAR-10 with a 20% reduction in Expected Calibration Error (ECE).	
Diabetic Retinopathy Detection System	Sep. 2024 – Nov. 2024
- Engineered a desktop analysis tool for retinal disease classification with Grad-CAM for interpretability. - Fine-tuned InceptionV3 on the APTOS dataset, using data augmentation to achieve 88% accuracy.	

LEADERSHIP & SERVICE

Board Member – General Secretary, Bagi Dunia (NGO)	Feb. 2024 – Present
<i>Jakarta, Indonesia</i> - Directed financial operations and fundraising, securing Rp17M+ (USD ≈\$1000) and seven corporate sponsorships. - Managed 30+ volunteers and coordinated logistics for food distribution aid to over 100 beneficiaries. - Supervised volunteering volumes, Treasury & Public Relations Operations.	
Academic Tutor	Nov. 2023 – Dec 2025
<i>Independent</i> <i>Jakarta, Indonesia</i> Conducted weekly tutorials in Statistics, Discrete Math, Computational Physics and Linear Algebra; organized technical bootcamps reaching 150+ students.	

CERTIFICATIONS

Course Certifications
NVIDIA Deep Learning Institute (Dec 2025): Building LLM Applications With Prompt Engineering Kaggle ML (2025): Computer Vision, Model Explainability, Deep Learning FreeCodeCamp (2024-2025): Scientific Computing (Python), Machine Learning (Python), Algorithms

TECHNICAL SKILLS

Languages: Python, C, SQL, JavaScript, Swift (learning), Java
Deep Learning: PyTorch, TensorFlow, Keras, Vision Transformers (ViT), CNNs, Object Detection, Transfer Learning
MLOps & Tools: Git, Weights & Biases (W&B), Docker (basic), CoreML (learning), NumPy, Pandas, OpenCV, Grad-CAM, DICOM
Research: Experimental Design, Ablation Studies, Statistical Analysis, Technical Writing, Data Augmentation & Pipeline curation, Model Calibration

LANGUAGES

English (Professional Proficiency)	Indonesian (Native)
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